

Robustness of ARloTEDEF against Adversarial Attacks in IoT Systems

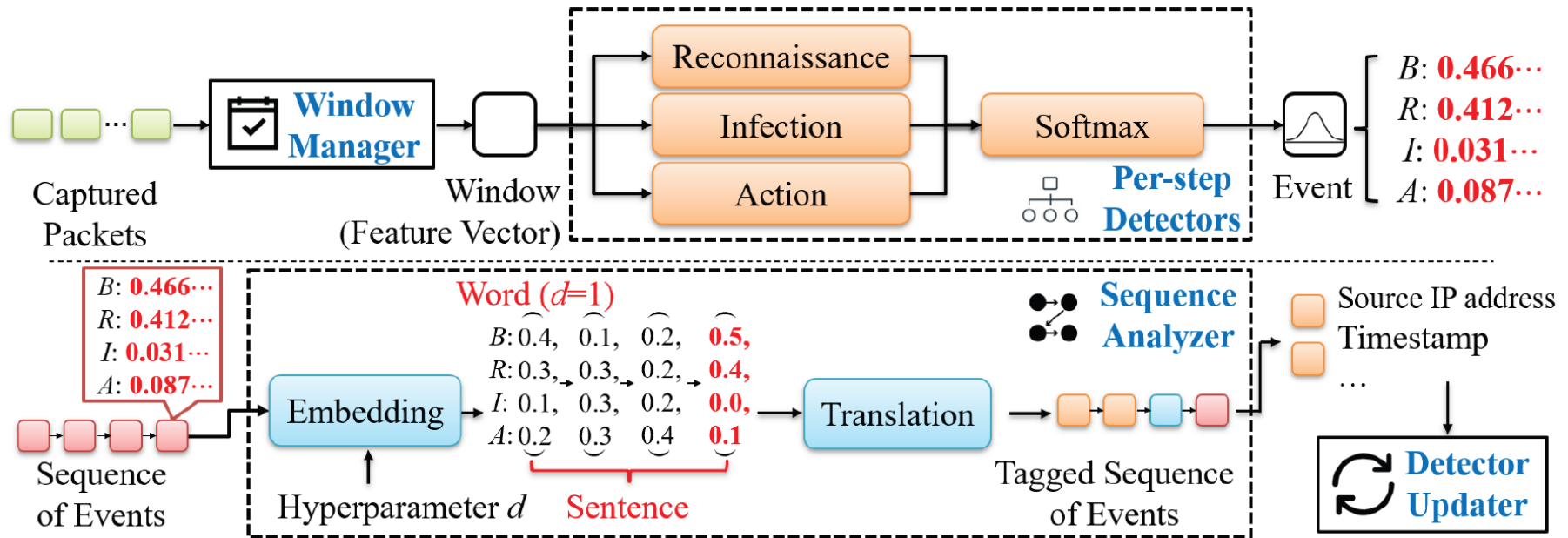
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Workflow of ARIoTEDef System



- Detect whether the **traffic within a window** belongs to three anomaly categories
- Generate **sequences of events** from probability distributions of several windows
- Use a **sequence-to-sequence** model to analyze whether each event is anomaly
 - The analysis in this module **leverages the attention mechanism** that combines information from contextual events
- Update the infection detector with **identified infection events** and benign events

Threat Model

● **Traffic Generated Based on The Black-box Attack Model**

➤ Attacker has no knowledge about the target system.

➤ Attack method:

□ Non-adversarial evasion

✓ Zero-day Attack

✓ Stealth Attack

□ **Adversarial evasion**

✓ Boundary

✓ HopSkipJump

- Has access to the predicted label of the model.
- Relies on input manipulation, query optimization, and exploration to understand the target model.

● **Traffic Generated Based on The White-box Attack Model**

➤ Attacker has complete knowledge about the target system.

➤ Attack method:

□ **Adversarial evasion**

✓ FGSM

✓ PGD

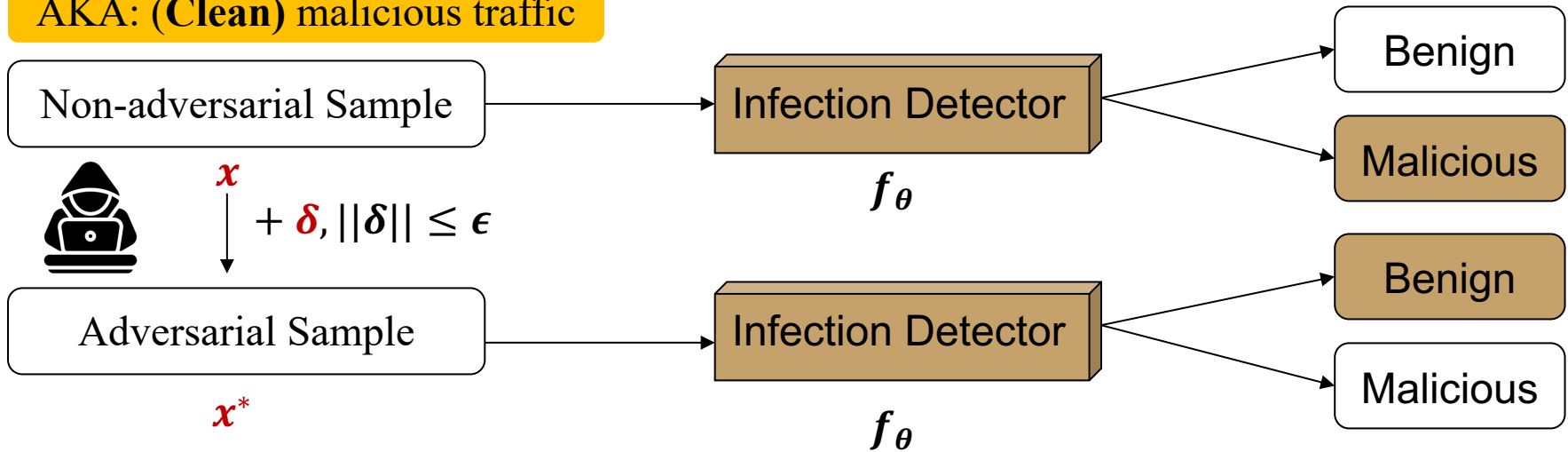
- Has access to the architecture, parameters, and internal workings of the model.
- Craft highly optimized adversarial samples.

Adaptability: Adversarial samples generalizes well across different models.

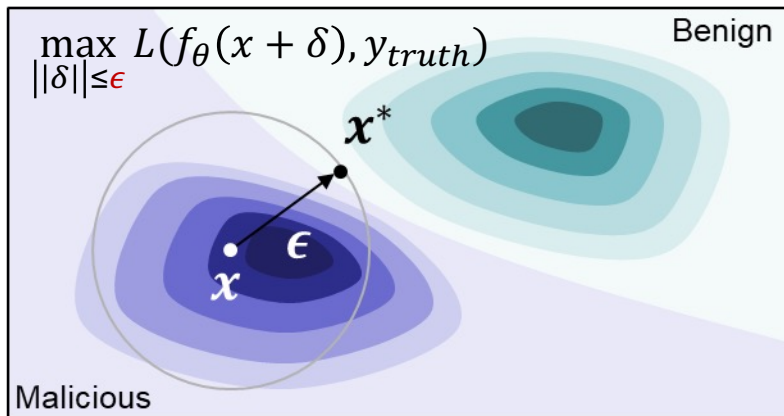
Success rate: Exploit vulnerabilities of the model, improve the evade success rate. 3

White-box Gradient-based Adversarial Sample

AKA: (**Clean**) malicious traffic



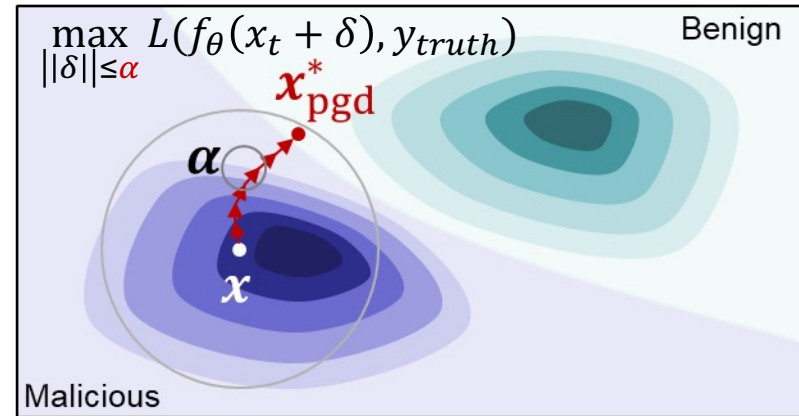
FGSM Attack



● Create perturbation in a **single step**

$$x^* = x + \epsilon \cdot \text{sign}(\nabla_x L(f_\theta(x), y_{truth}))$$

PGD Attack

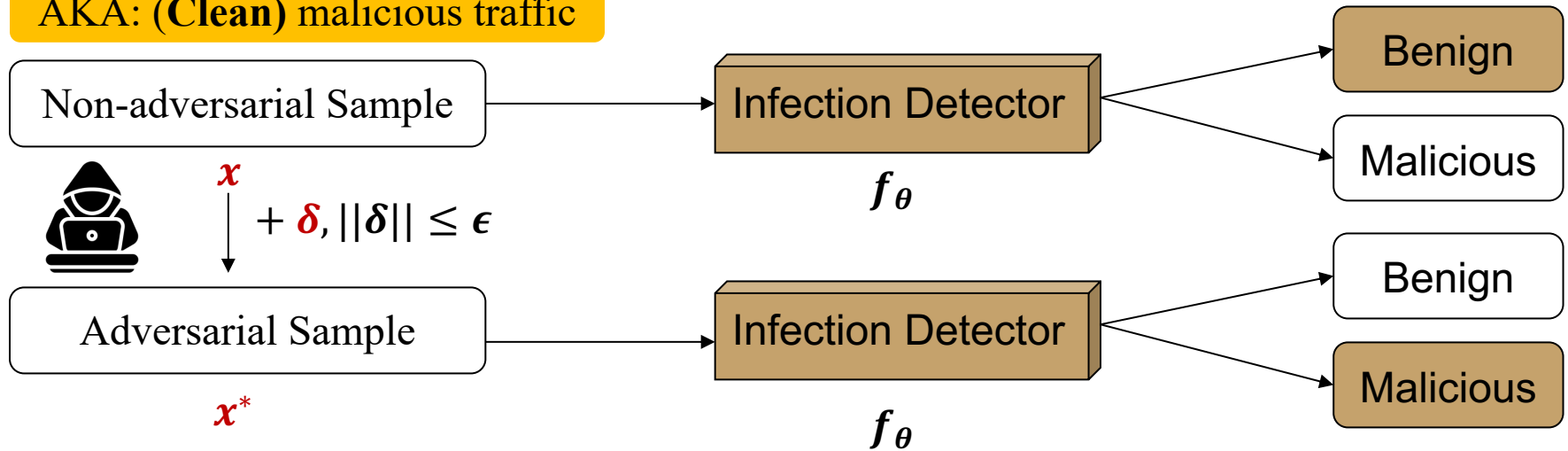


● Iteratively create perturbation

$$x^{t+1} = \text{Clip}_\epsilon(x^t + \alpha \cdot \text{sign}(\nabla_x L(f_\theta(x_t), y_{truth}))) \quad 4$$

White-box Gradient-based Adversarial Sample

AKA: (Clean) malicious traffic

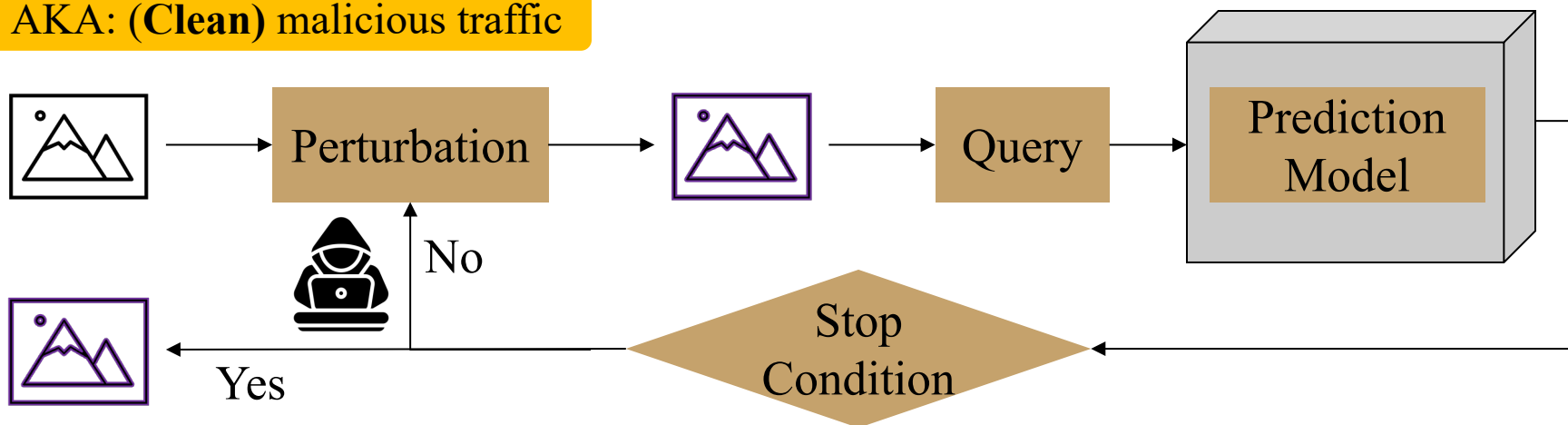


Performance of Pre-evolved Detectors on **White-box Adversarial Attacks**

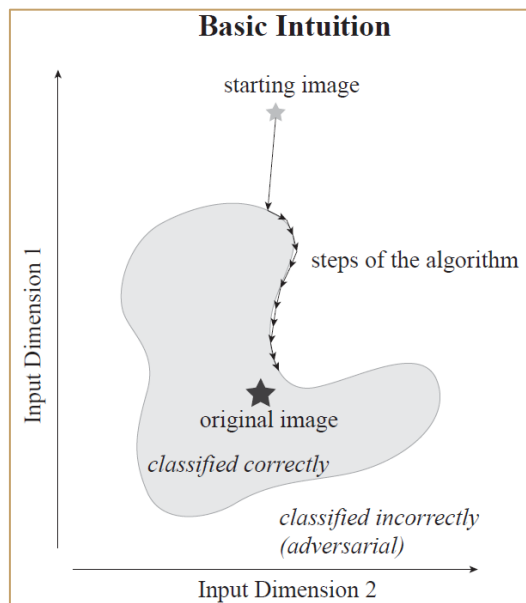
Optimize	Input				Reconnaissance Detector				Infection Detector			
	Name	ϵ	ϵ_{ite}	n_{ite}	TP	FN	FNR(%)	Recall(%)	TP	FN	FNR(%)	Recall(%)
-	Clean	-	-	-	3064	13	0.42	99.58	126	192	60.38	39.62
Untargeted	FGSM	0.5	-	-	66	3011	97.86	2.14	5	313	98.43	1.57
	FGSM	1.0	-	-	48	3029	98.44	1.56	0	318	100.00	0.00
	PGD	0.5	0.25	20	48	3029	98.44	1.56	5	313	98.43	1.57
	PGD	1.0	0.5	20	29	3048	99.06	0.94	0	318	100.00	0.00
Targeted	FGSM	0.5	-	-	67	3010	97.82	2.18	110	208	65.41	34.59
	FGSM	1.0	-	-	52	3025	98.31	1.69	96	222	69.81	30.19
	PGD	0.5	0.25	20	89	2988	97.11	2.89	100	218	68.55	31.45
	PGD	1.0	0.5	20	239	28.38	92.23	7.77	4	314	98.74	1.26

Black-box Decision-based Adversarial Sample

AKA: **(Clean)** malicious traffic

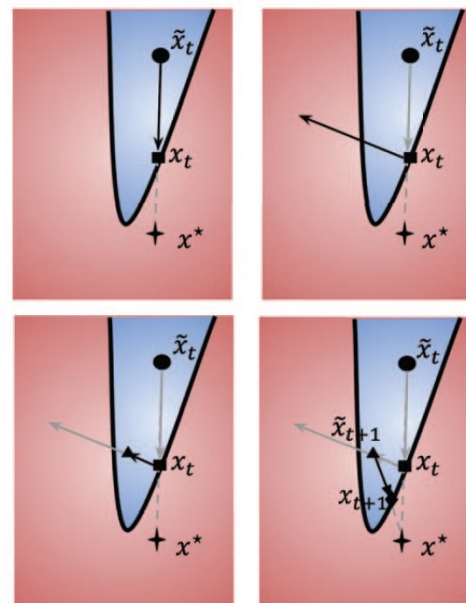


Boundary Attack [ICLR18]



- Starts from a large adversarial perturbation
- Search boundary search via a binary search
- Seeks to reduce the perturbation while staying adversarial.

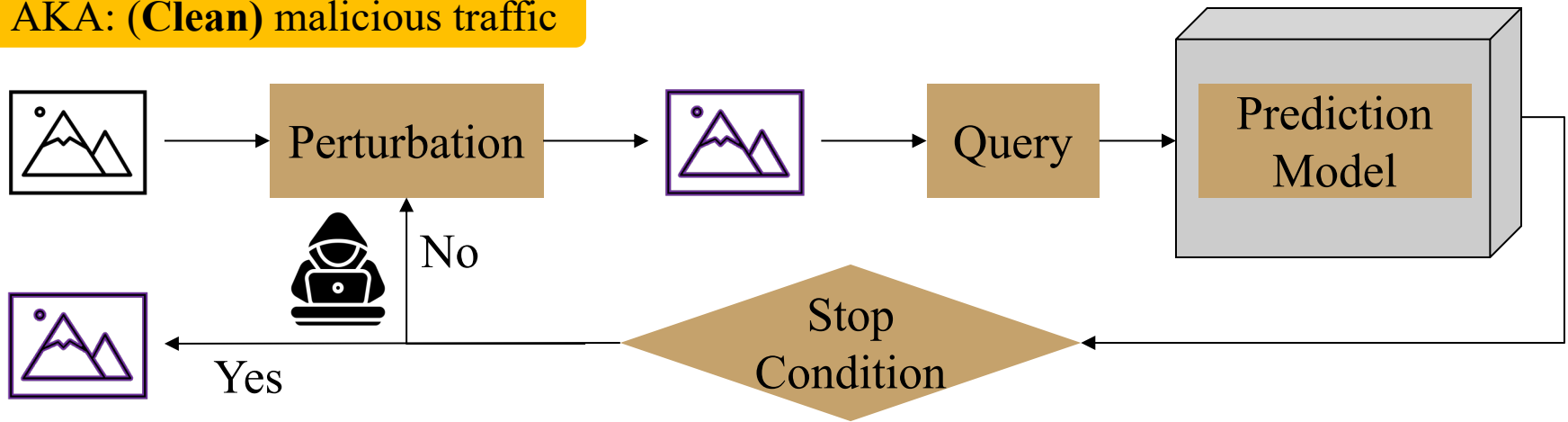
HopSkipJump Attack [S&P20]



- Estimate gradient direction
- Search step-size (Hop/Skip/Jump benefit wider search area)
- Search boundary via a binary search

Black-box Decision-based Adversarial Sample

AKA: (**Clean**) malicious traffic

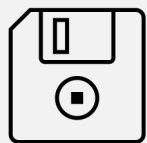


Performance of Pre-evolved Detectors on **Black-box Adversarial Attacks**

Input	ϵ	IterNum	QueryNum	Reconnanssance Detector			Infection Detector			Action Detector		
				TP	FN	Recall (%)	TP	FN	Recall (%)	TP	FN	Recall (%)
Clean	-	-	-	3064	13	99.58	126	192	39.62%	112	11	91.06
Boundary	1.0	500000	100000	3064	13	99.58	126	192	39.62%	112	11	91.06
HopSkipJump	-	500000	100000	3064	13	99.58	126	192	39.62%	112	11	91.06

Attack and Defense Settings

● Adversarial Attack and Defense Setting **I** (N rounds of Evolution)



Clean Infection Dataset $X = \{x_{benign}, x_{malicious}\}$

Construct adversarial malicious samples based on the latest infection detector from clean malicious samples



M

Infection Detector



Adversarial Infection Dataset $X_{adv} = \{x_{malicious}^*\}$



$X_{adv_1} \leftarrow X + M_0$



X_{adv_1}



$M_0 \rightarrow M_1$



$X_{adv_2} \leftarrow X + M_1$

X_{adv_2}

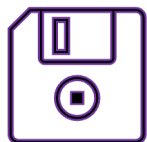
$M_1 \rightarrow M_2$



$\vdots$

$\vdots$

$\vdots$



$X_{adv_N} \leftarrow X + M_{N-1}$

X_{adv_N}

$M_{N-1} \rightarrow M_N$



Research Questions

- **Q1: Does the number of False Negatives (FN) of the detector **decrease** after several rounds of evolution?**
 - As the number of evolution rounds increases, the detector is retrained by more and more positive samples, especially some adversarial malicious samples that escaped the detector. So, the evolved detector will be familiar with these infection patterns and more likely to predict them as malicious. Thus, FN will decrease.
- **Q2: Does the number of False Positives (FP) of the detector **increase** after several rounds of evolution?**
 - As the number of evolution rounds increases, it is easier to predict negative (benign) samples as positive samples while TP increases. Because the adjustment of the decision boundary resulted from evolution may lead to the increase of FP while reducing FN.

$$FN\ Rate = \frac{FN \downarrow}{P = TP + FN}$$

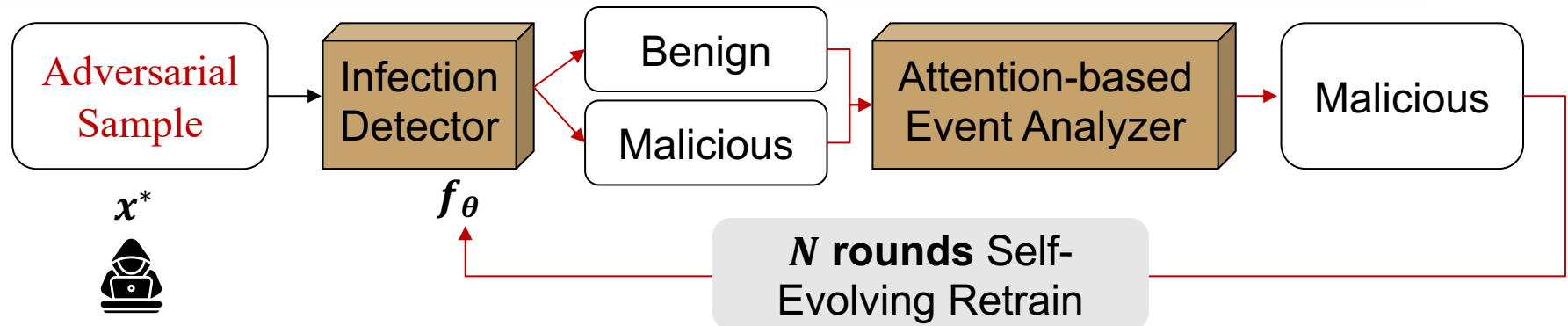
$$FP\ Rate = \frac{FP \uparrow}{N = TN + FP}$$

$$Recall = \frac{TP}{TP + FN \downarrow}$$

$$Precision = \frac{TP}{TP + FP \uparrow}$$

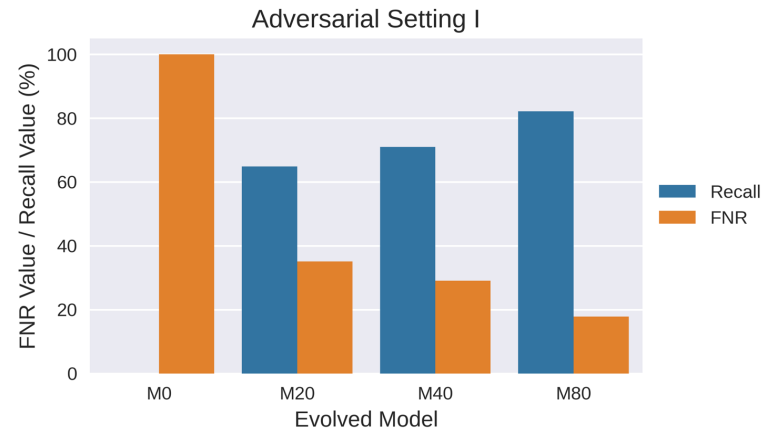
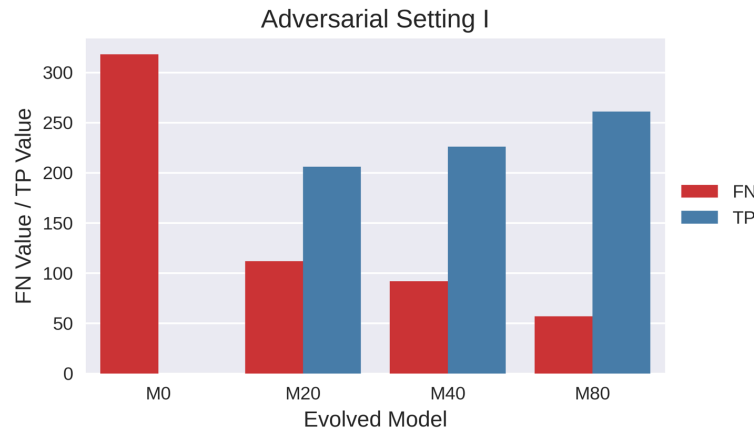
$$F1\ score = 2 * \frac{Precision * Recall}{Precision + Recall}$$

Adversarial Setting I

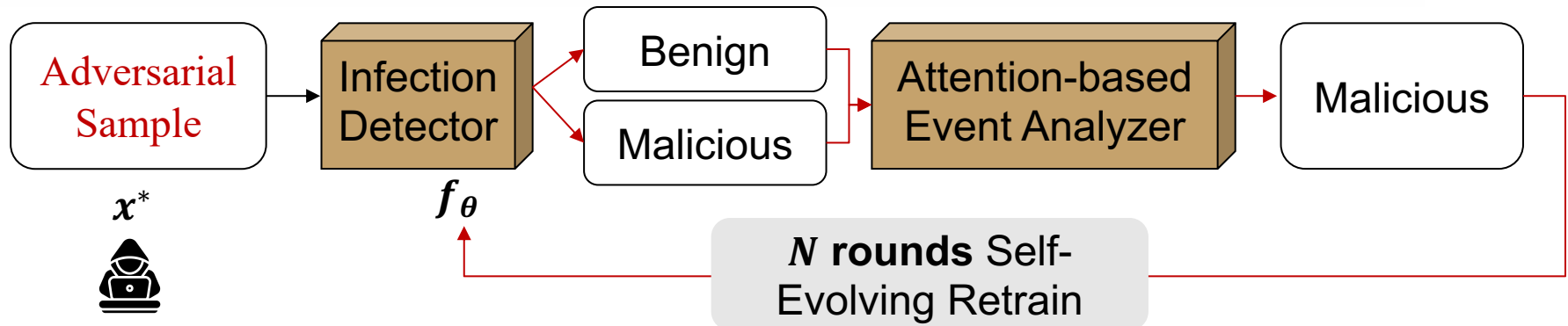


Performance of the Evolved Infection Detector on Adversarial Samples

Evolution Round	0	20				40				80			
Seed	-	0	1	2	average	0	1	2	average	0	1	2	average
TP	0	194	198	227	206	232	233	212	226	229	305	<u>250</u>	261
FN	318	124	120	91	112	86	85	106	92	89	13	<u>68</u>	57
Q1 FNR(%)	100.00	38.99	37.74	28.62	35.12	27.04	26.73	33.33	29.04	27.99	4.09	<u>21.38</u>	17.82
Recall(%)	0.00	61.01	62.26	71.38	64.88	72.96	73.27	66.67	70.96	72.01	95.91	<u>78.62</u>	82.18

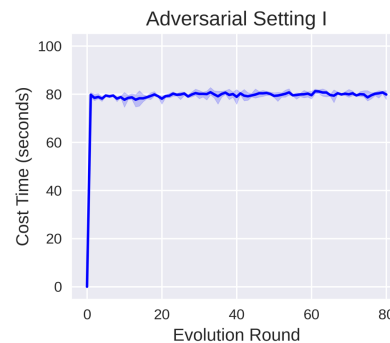
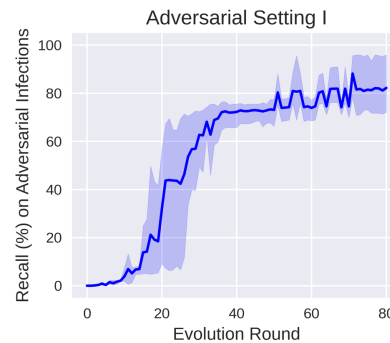
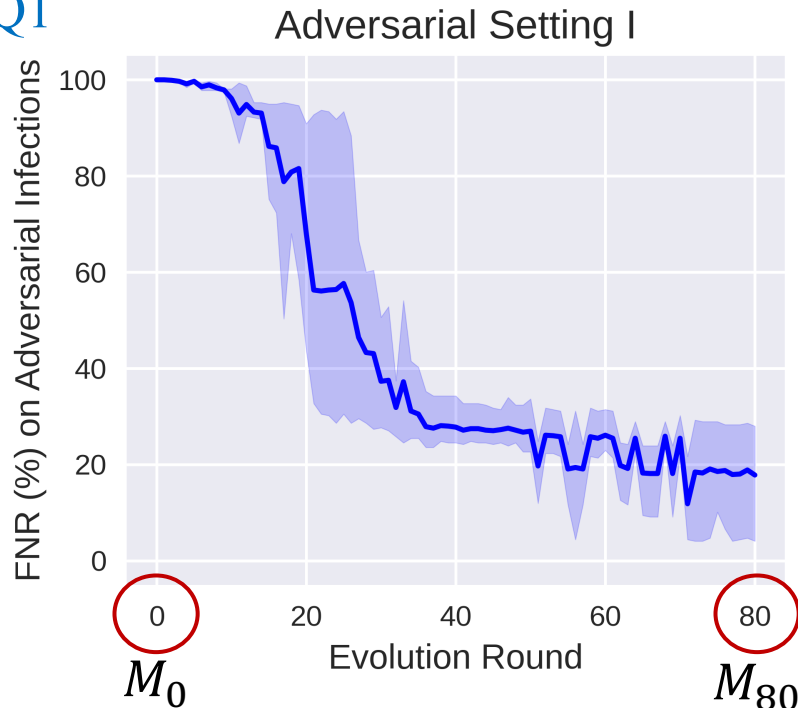


Adversarial Setting I



Impact of the Number of Evolution Rounds on Adversarial Robustness

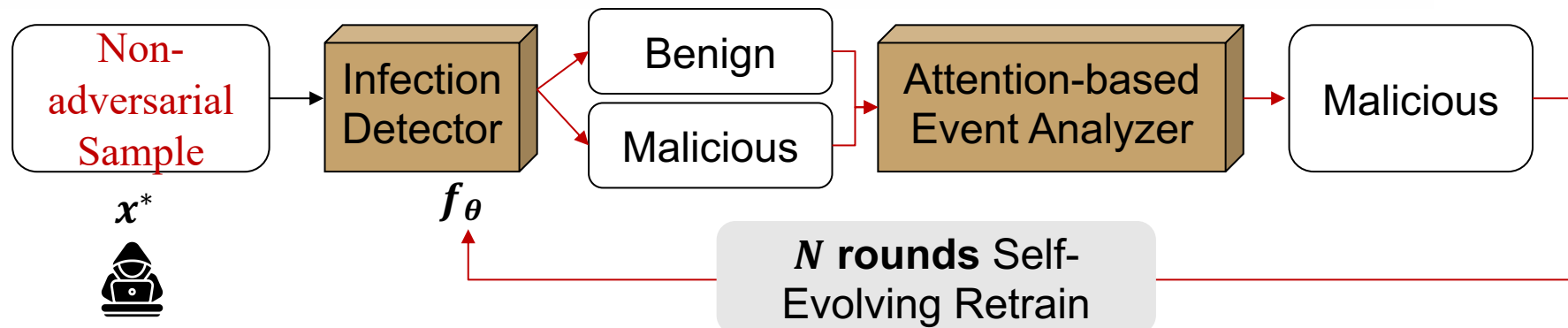
Q1



Recall increased significantly.

Within 30 rounds of evolution, the accuracy of the detector on white-box adversarial samples can increase to more than 50%.

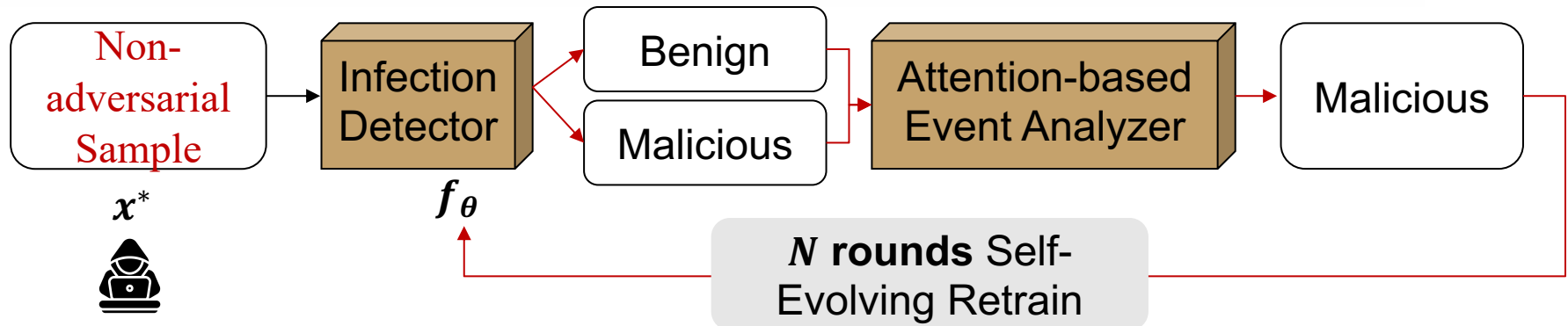
Adversarial Setting I



Performance of Evolved Infection Detector on The **Clean** Test Set

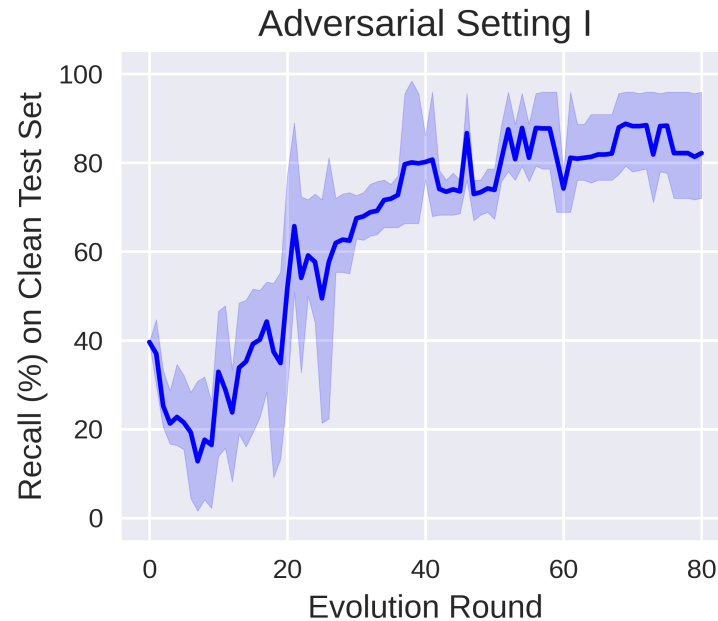
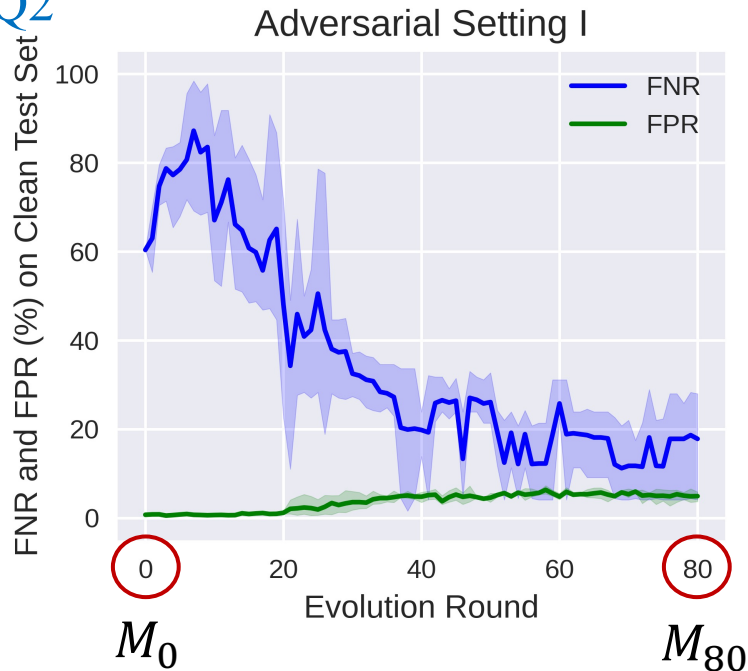
Evolution Round	Seed	TP	FN	TN	FP	FNR(%)	FPR(%)	Precision(%)	Recall(%)	F1(%)	Accuracy(%)
0	-	126	192	3887	28	60.38	0.72	81.82	39.62	53.39	94.80
20	0	215	103	<u>3793</u>	<u>122</u>	32.39	<u>3.12</u>	<u>63.80</u>	67.61	65.65	94.68
	1	214	104	3766	149	32.70	3.81	58.95	67.30	62.85	94.02
	2	<u>292</u>	<u>26</u>	3719	196	<u>8.18</u>	5.01	59.84	<u>91.82</u>	<u>72.46</u>	94.76
	average	240	78	3759	156	24.42	3.98	60.86	75.58	66.98	94.49
40	0	238	80	3676	239	25.16	6.10	49.90	74.84	59.87	92.46
	1	234	84	3671	244	26.42	6.23	48.95	73.58	58.79	92.25
	2	214	104	3783	132	32.70	3.37	61.85	67.30	64.46	94.42
	average	229	89	3710	205	28.09	5.24	53.57	71.91	61.04	93.05
80	0	229	89	3685	230	27.99	5.87	49.89	72.01	58.94	92.46
	1	305	13	3712	203	4.09	5.19	60.04	95.91	73.85	<u>94.90</u>
	2	250	68	3770	145	21.38	3.70	63.29	78.62	70.13	94.97
	average	261	57	3722	193	17.82	4.92	57.74	82.18	67.64	94.11

Adversarial Setting I



Performance of Evolved Infection Detector on The **Clean** Test Set

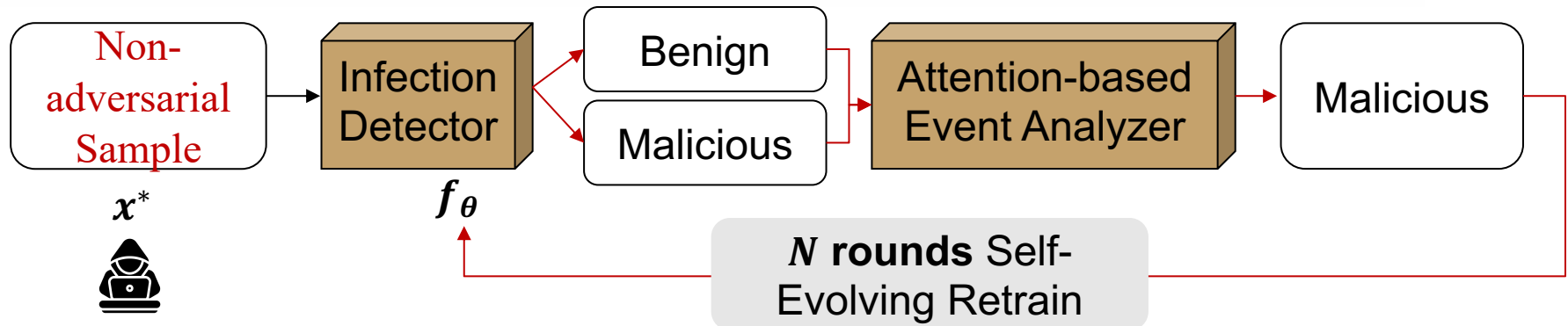
Q2



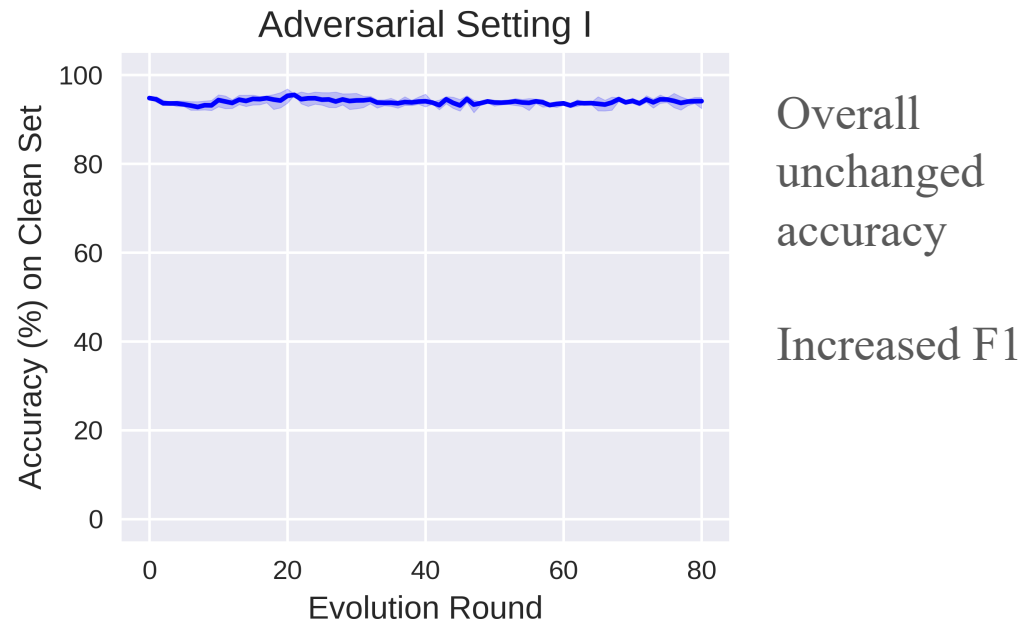
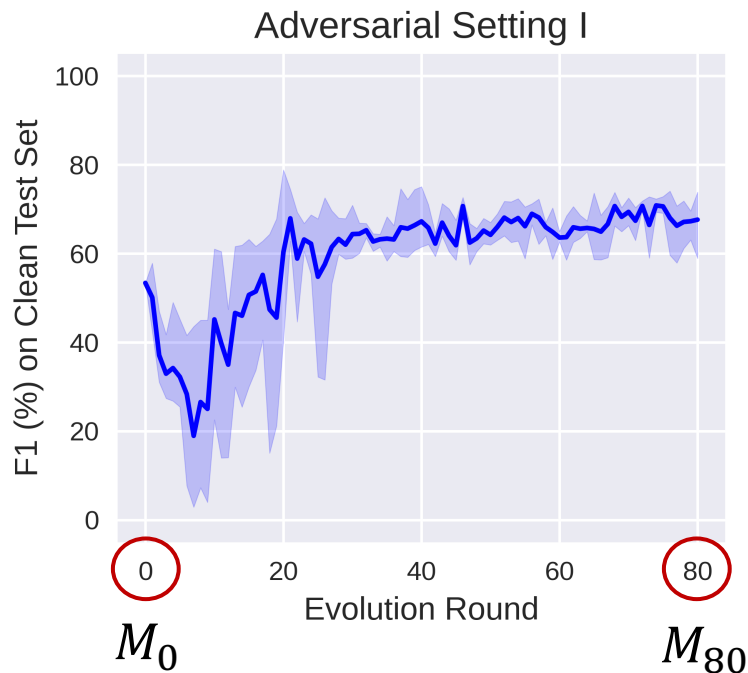
Recall increased significantly.

Sharp drop of FNR, weak increase of FPR

Adversarial Setting I

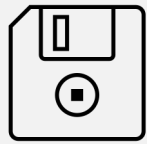


Performance of Evolved Infection Detector on The **Clean** Test Set



Attack and Defense Settings

● Adversarial Attack and Defense Setting II (N rounds of Evolution)



Clean Infection Dataset $X = \{x_{benign}, x_{malicious}\}$



...



...



Clean Infection Dataset X_1

Clean Infection Dataset X_{N+1}



M

Infection Detector



Adversarial Infection Dataset $X_{adv1} \leftarrow X_1 + M_0$

Adversarial Infection Dataset $X_{advN+1} \leftarrow X_{N+1} + M_N$



$X_{adv1} \leftarrow X_1 + M_0$



X_{adv1}



$M_0 \rightarrow M_1$



$X_{adv2} \leftarrow X_2 + M_1$

X_{adv2}

$M_1 \rightarrow M_2$



...

...

...



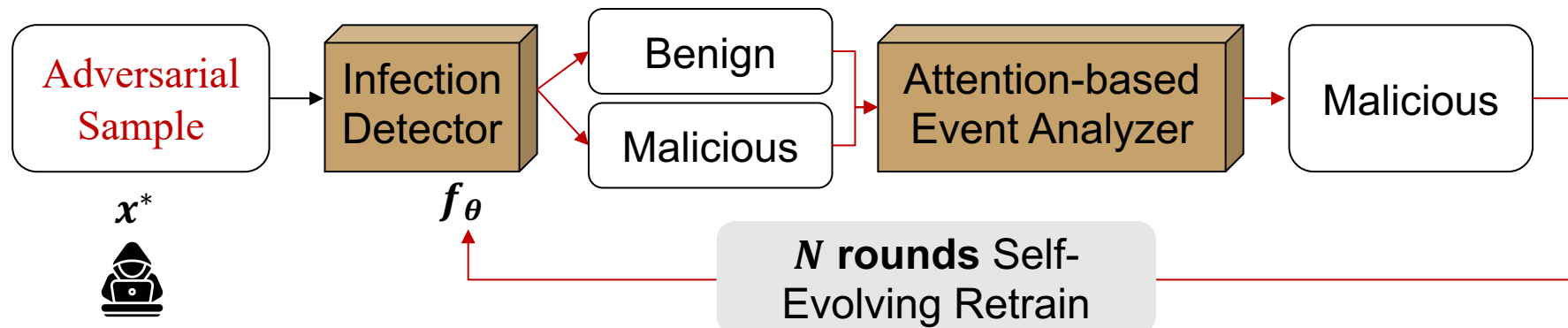
$X_{advN} \leftarrow X_N + M_{N-1}$

X_{advN}

$M_{N-1} \rightarrow M_N$

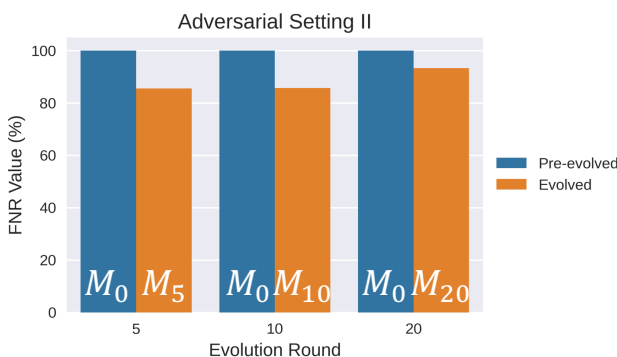
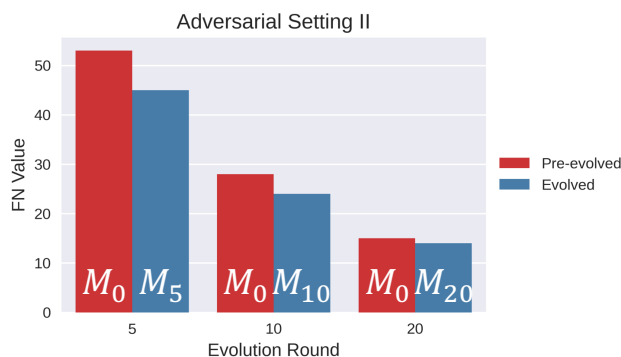


Adversarial Setting II



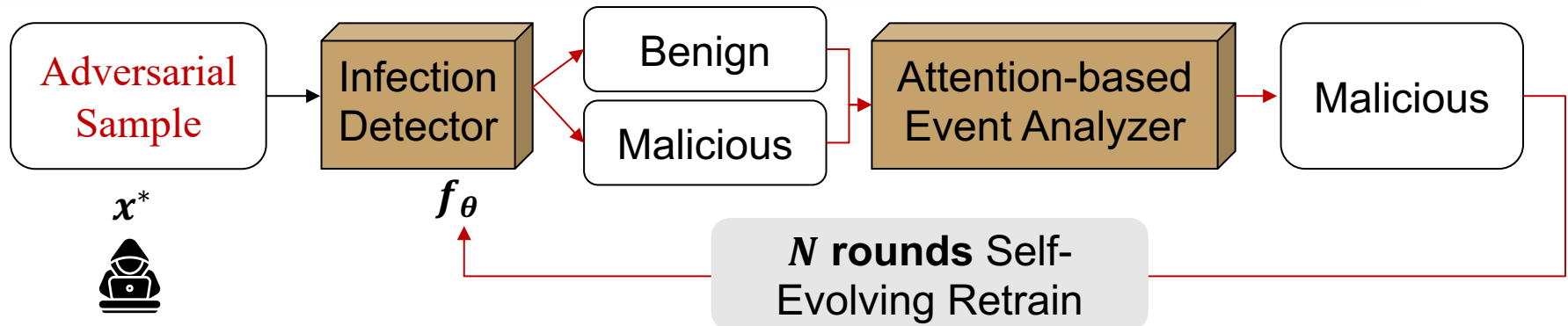
Performance of the Evolved Infection Detector on Adversarial Samples

AdvSet Size	53					28					15				
Evolution Round	0	5				0	10				0	20			
Seed	-	0	1	2	average	-	0	1	2	average	-	0	1	2	average
TP	0	3	17	3	8	0	6	6	0	4	0	1	1	1	1
FN	53	50	36	50	45	28	22	27	28	24	15	14	14	14	14
FNR(%)	100.00	94.34	67.92	94.34	85.53	100.00	<u>78.57</u>	<u>78.57</u>	100.00	85.71	100.00	93.33	93.33	93.33	93.33
Recall(%)	0.00	5.66	32.08	5.66	14.47	0.00	<u>21.43</u>	<u>21.43</u>	0.00	14.29	0.00	6.67	6.67	6.67	6.67

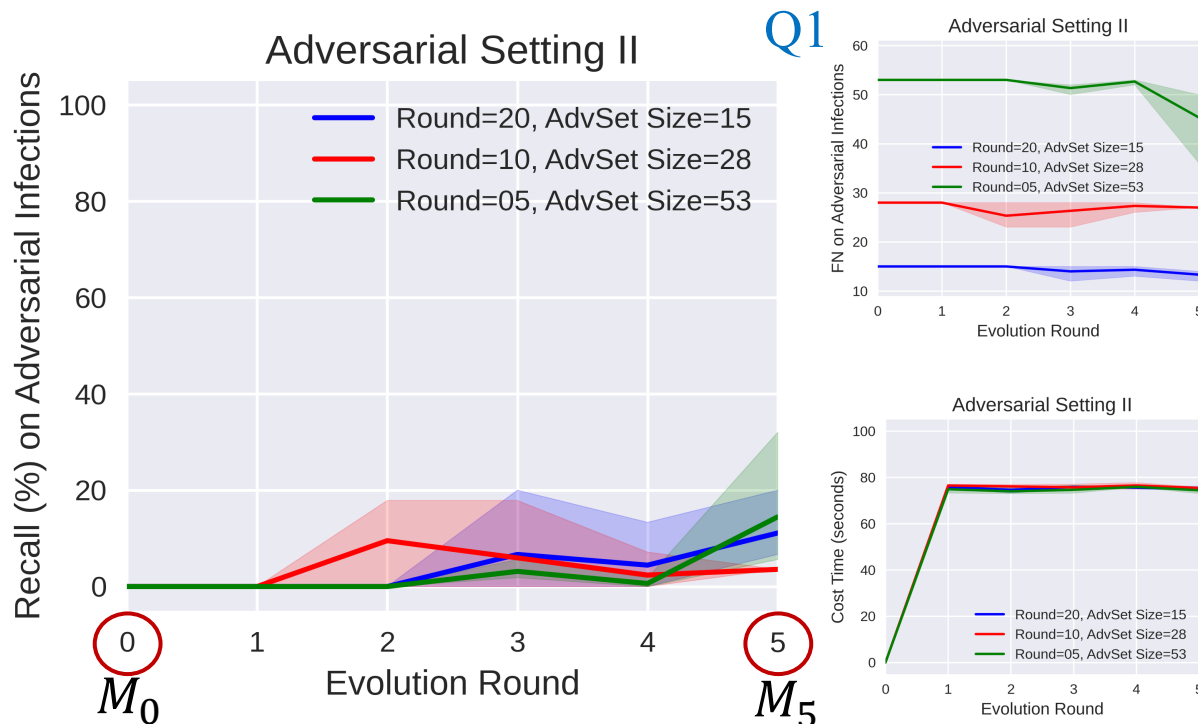


Robustness improvement by self-evolution is observed under different AdvSet sizes.

Adversarial Setting II

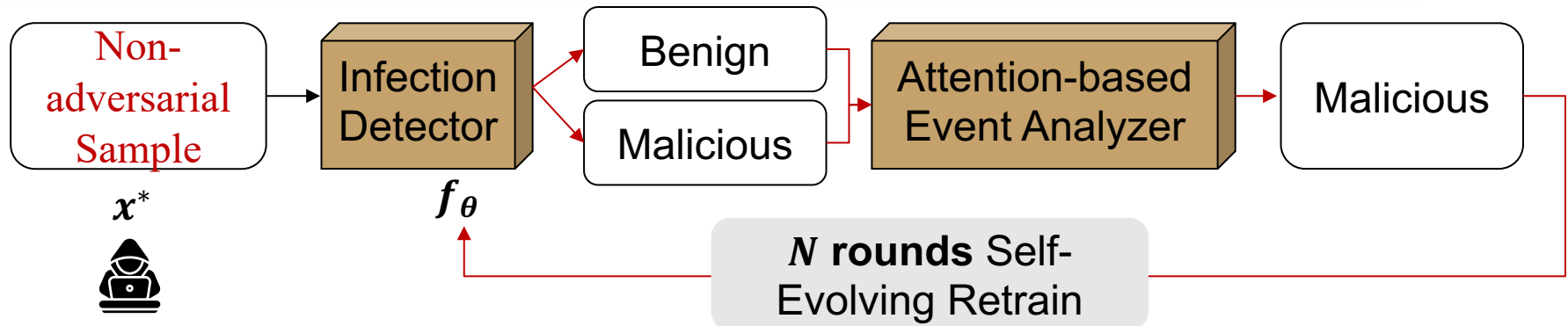


Impact of the Number of Adversarial Samples on Adversarial Robustness



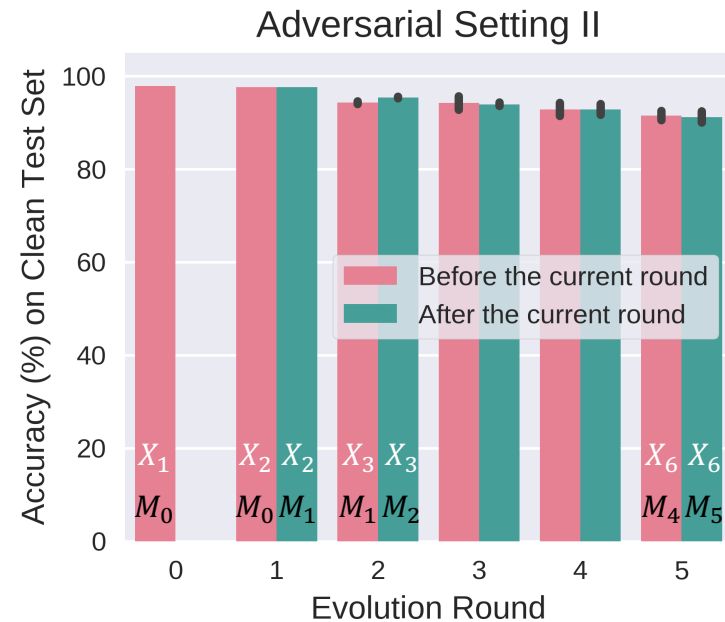
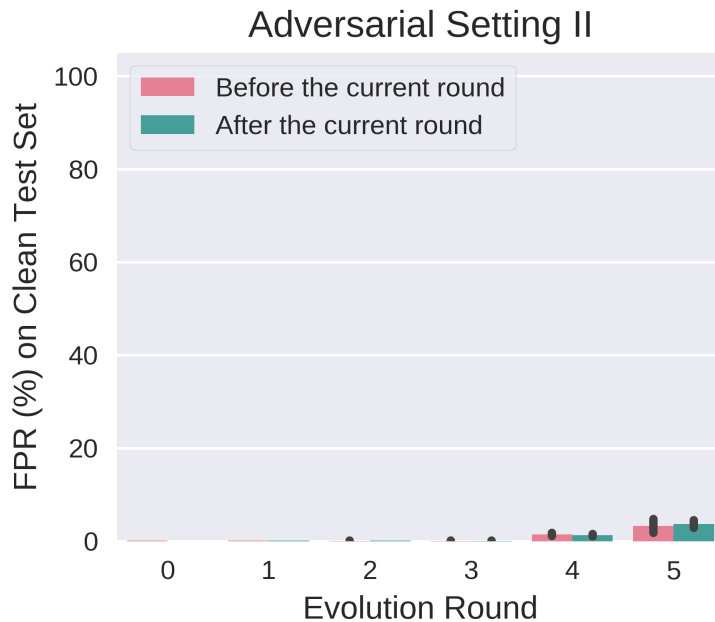
No matter whether the attack is fast or slow (more or less adversarial samples are sent in a certain period of time), the self-evolution strategy can increase recall, reflecting enhanced robustness.

Adversarial Setting I

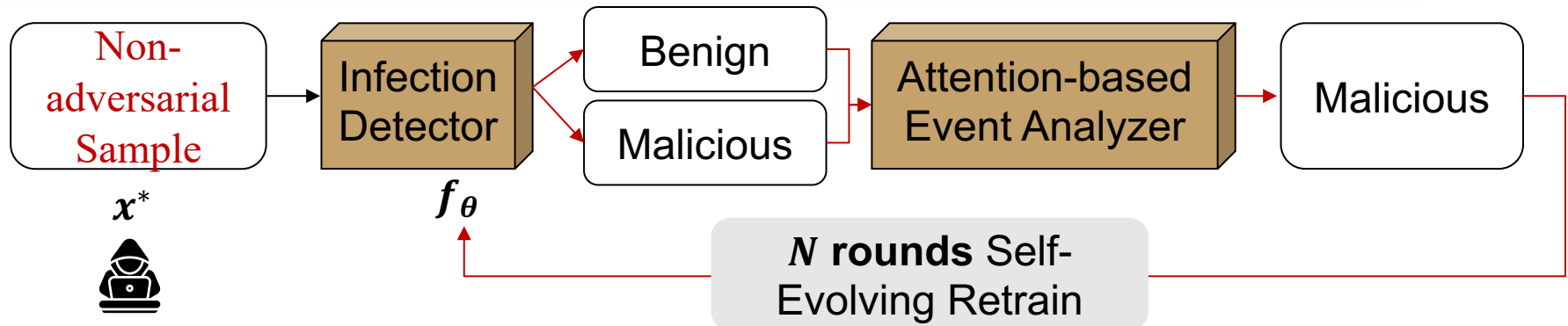


Performance of Evolved Infection Detector on The **Clean** Test Set

Q2

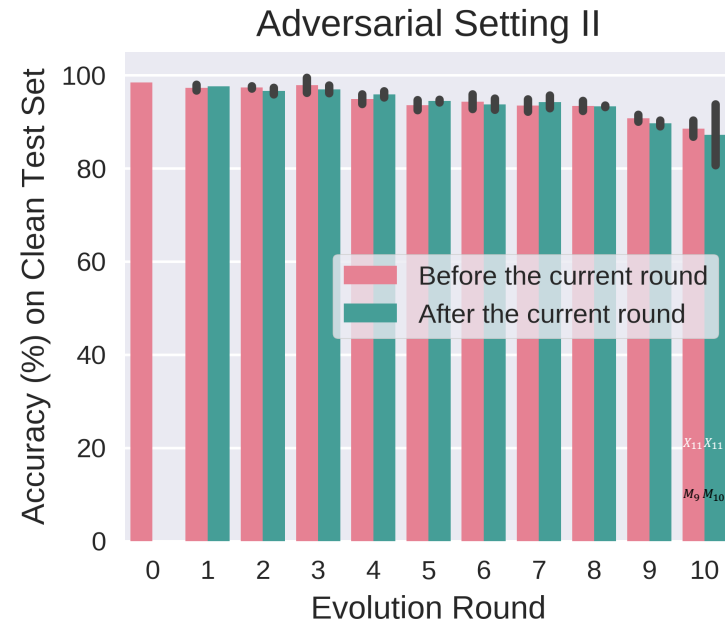
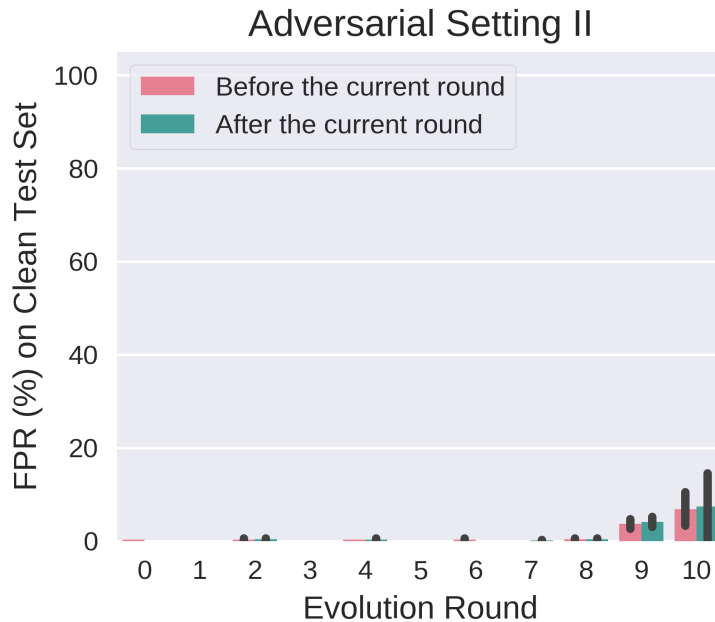


Adversarial Setting I

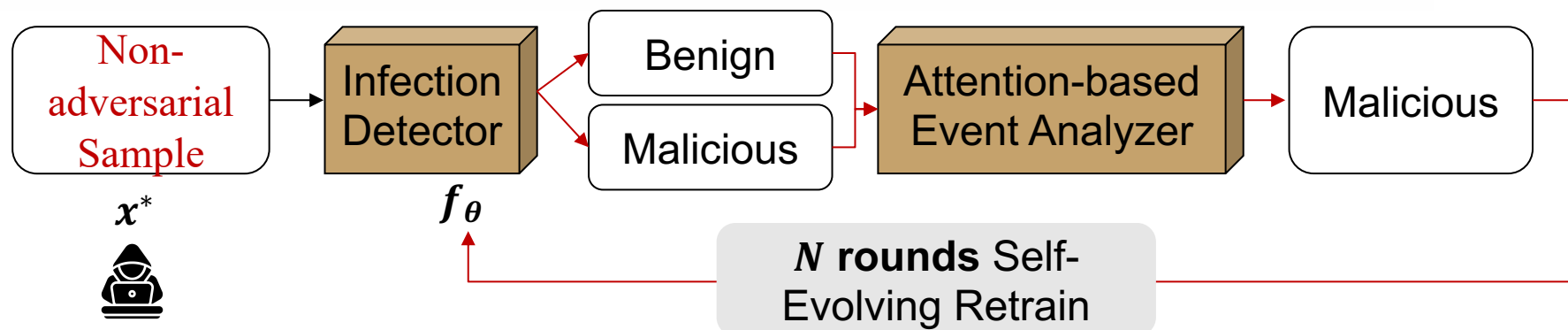


Performance of Evolved Infection Detector on The **Clean** Test Set

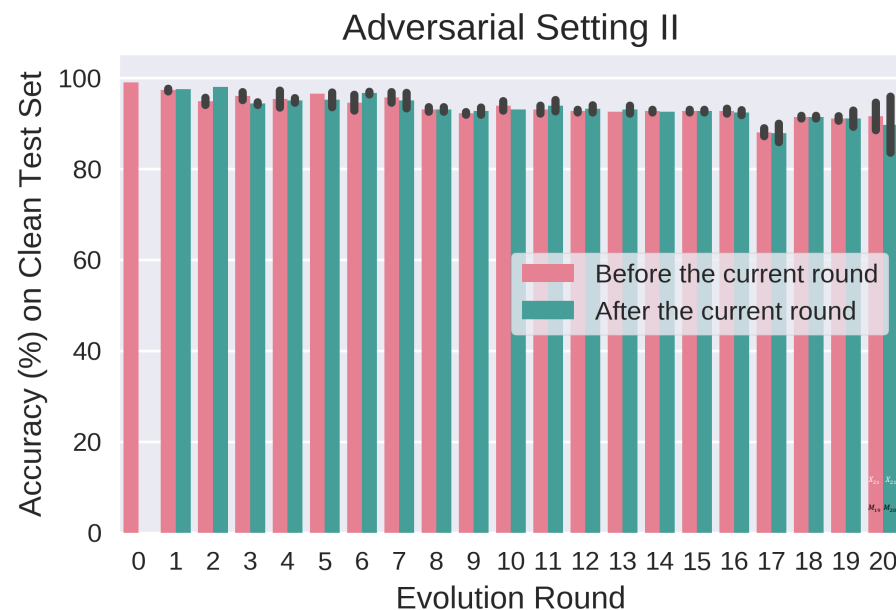
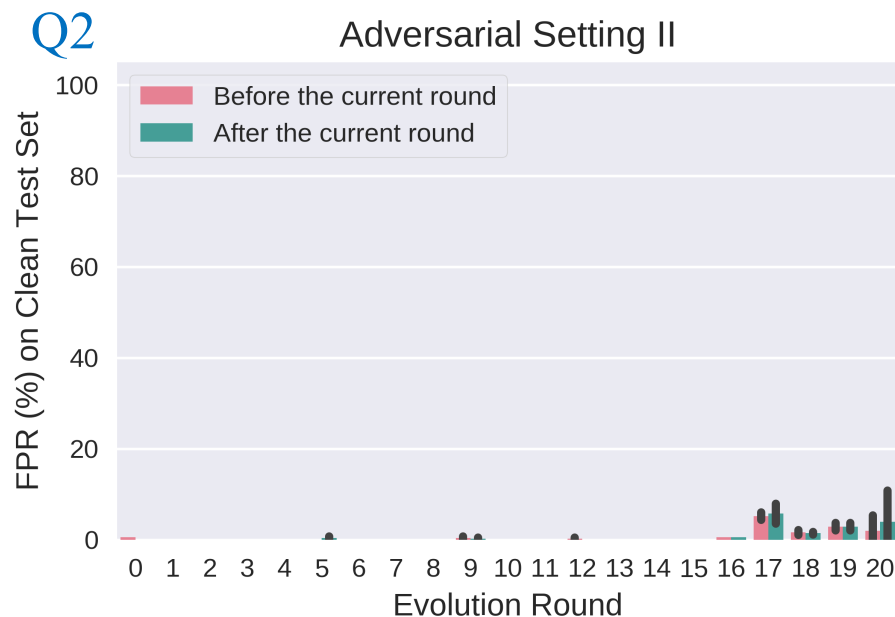
Q2



Adversarial Setting I



Performance of Evolved Infection Detector on The **Clean** Test Set



Conclusion

C1: After multiple rounds of evolution, the robustness of the infection detector on white-box adversarial evasion attacks improves.

Taking 80 rounds of evolution as an example, the average FN rate drops from 100% to 17%.

C2: After multiple rounds of evolution, the robustness of the infection detector on black-box non-adversarial evasion attacks also improves.

Taking round 80 as an example, the FN rate drops from 60% to 17%, and the FP rate has only a slight increase, from 0.7% to 4.9%.

C3: The Pre-evolved Action detector is very robust to adversarial white-box and adversarial black-box attacks, and it is difficult for an attacker to mislead it.

Four adversarial evasion attacks (PGD\FGSM\Boundary\HopSkipJump) have almost 0% success rate.

Thank You

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